

2024



US Army Provides Anywhere-to-Anywhere Care in Combat Situations Using AWS Wickr

Learn how AWS Partner Deloitte helped the US Army Telemedicine & Advanced Technology Research Center (TATRC) facilitate secure collaboration between medical teams using AWS Wickr.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#)

Over 60

hospitals empowered with telehealth

15 seconds

for trainees to adopt the solution

Hundreds

of lives saved



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Overview

The [US Army Telemedicine and Advanced](#) 1 through the National Emergency Tele-Criti

Wickr and AWS Private 5G. During the COVID-19 pandemic, TATRC needed to help hospitals that were coping with limited resources connect with medical specialists. TATRC built NETCCN to address this challenge and provide telemedicine in combat and training scenarios, using AWS Wickr to comply with Depa



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2

(DoD) requirements. This solution empowered TATRC to deliver care virtually anywhere in response to pandemics, natural disasters, war, and other acute events.



Opportunity | Supporting Patients in Critical Condition with Telemedicine

Based in Fort Detrick (Frederick, Maryland), TATRC is a research organization that addresses critical gaps in healthcare. The organization knows that, when it comes to medical care, every second counts. At the onset of the COVID-19 pandemic, hospitals and healthcare systems faced unprecedented challenges. Many areas, especially rural ones, did not have the resources needed to treat severely sick patients, and they faced a lack of intensive care unit beds and medical specialists.

After obtaining research funding, TATRC was tasked with creating a telemedicine solution that would connect qualified medical personnel to healthcare workers who needed help treating patients in critical condition. “We needed a way to deliver clinical expertise to where it is needed, and to do it in a way that was as simple as downloading an app and using it,” says Matt Quinn, science director at TATRC. That’s where Amazon Web Services (AWS) came in.

TATRC engaged AWS Partner [Deloitte](#) to help develop NETCCN: a low-resource, cloud-based, virtual health solution that facilitates secure collaboration between onsite and remote medical teams. Given the sensitive nature of the medical data needing to be transmitted and stored, the solution had to not only meet DoD security requirements but also comply with HIPAA regulations. Based on these needs, Deloitte recommended that TATRC use [Wickr](#), a service for protecting one-to-one and group messaging, voice and video calling, file sharing, and location sharing with end-to-end encryption.



With its advanced security, administrative controls, and data retention capabilities, AWS Wickr was an ideal choice for TATRC. Furthermore, this AWS service had an existing authorization to operate on DoD networks. “The DoD works at the pace of authorizations, and it could take a year or longer to get authorization to operate on a network,” says Quinn. “The AWS Wickr team had the foresight to invest in DoD authorization, which saved us years of waiting and millions of dollars.”

Solution | Expanding NETCCN to Military Contexts

During the initial COVID-19 pandemic response, NETCCN was successfully deployed in over 60 hospitals across the United States, with a hospital in Missouri going live with the solution in 3 hours. Following the success of this project, TATRC deployed NETCCN in response to a COVID-19 surge in Guam, where it played a critical, lifesaving role. The island, located in the Pacific Ocean, is relatively isolated and geographically distant from outside support. The US Navy operates a hospital on the island as part of an emergency response network; during the COVID-19 pandemic, however, healthcare providers in Guam needed even more support. TATRC stepped in to provide critical telemedicine assistance, connecting local medical teams with experts from Naval Medical Center San Diego.

“The second day that we were live with NETCCN support, a nurse used it to save a patient’s life,” says Quinn. “She was the attending physician, and one of her patients became unstable due to tension pneumothorax, which is a deadly condition. Through NETCCN, she connected with an intensive care doctor in San Diego, who walked her through diagnosis and treatment. Together, they saved that patient.”

During the COVID-19 pandemic, TATRC noticed parallels between the shortage of intensive care unit beds and combat situations, in which casualties often exceed available medical resources. To better support medics in the field, it adapted the NETCCN framework for use in military contexts, renaming the solution the Military Emergency Tele-Critical Care Platform (METCC-P).

To use NETCCN for military operations, TATRC needed to adhere to DoD Impact Level 5 (IL5) guidelines, which cover the protection of “controlled unclassified information” transmitted over cloud services. Using its experience in combat casualty scenarios, Deloitte helped TATRC configure [AWS Wickr RAM](#) (Recall, Alert, and Messaging)—a version of AWS Wickr built for the DoD in collaboration with US Air Force Special Operations Command (AFSOC) and the General Dynamics Information Technology (GDIT) ARMA team—to meet IL5 standards. TATRC also adopted [AWS Private 5G](#), a managed service that makes it simpler to deploy, operate, and scale a private network, to extend METCC-P’s functionality in austere combat settings, which often have limited connectivity. This is crucial so

that, even in the most remote or compromised locations, military personnel can access and provide specialized medical care.

Outcome | Supporting More Patients and Providers with NETCCN/METCC-P

With NETCCN, TATRC has empowered medics on the ground to deliver lifesaving care with the help of encrypted video calls, messaging, and file sharing with medical professionals. It estimates that this solution has helped save hundreds of lives.

NETCCN/METCC-P has been used to train military medical students on how to deliver treatment in realistic combat scenarios. These students learned how to use the service in under 15 seconds, highlighting its simplicity of use—which is critical in combat scenarios. “With METCC-P, combat medics and nurses can now operate above their training,” says Quinn. “One user described it as being like having an intensive care doctor in her pocket.”

Using the NETCCN framework, TATRC can deliver care virtually anywhere in response to pandemics, natural disasters, war, and other acute, life-threatening events. The organization plans to expand NETCCN to more use cases in the future, adapting to new challenges and opportunities as they arise. The solution will soon be used for the Test to Treat initiative, a federal government project that aims to help people across the country access lifesaving treatments for COVID-19 at little to no cost.

“We’ve used NETCCN in so many ways that we didn’t envision when we started this project,” says Quinn. “There are so many other ways that we can use a simple, secure messaging platform. We are excited to expand the capability and capacity of the system to help more people.”

About US Army Telemedicine and Advanced Technology Research Center (TATRC)

The US Army Telemedicine and Advanced Technology Research Center (TATRC) is a military health research organization that addresses gaps in the Department of Defense medical program and military healthcare.

AWS Services Used

AWS Wickr

AWS Wickr is an end-to-end encrypted messaging and collaboration service with features designed to help keep your internal and external communications secure, private, and compliant.

[Learn more »](#)



AWS Private 5G

AWS Private 5G is a managed service that makes it easier to deploy, operate, and scale your own private mobile network, with all required hardware and software provided by AWS.

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2

